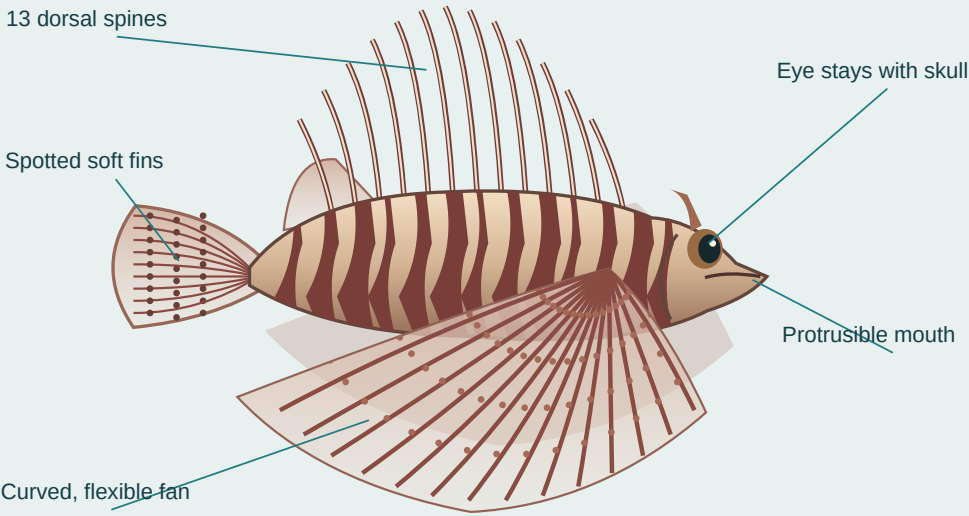


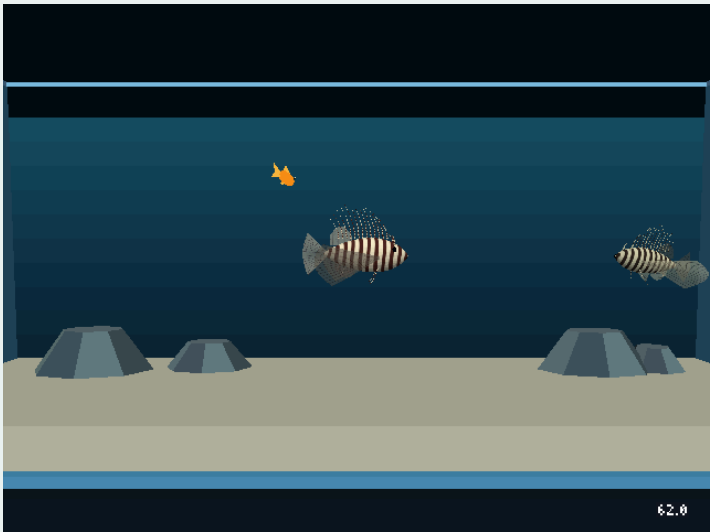
LIONFISH

Quiet fins. Sudden suction.

Pterois volitans · Shape, surface and motion in the game asset

RESEARCH = cited observations PROPOSAL = game design SCHEMATIC = original drawing





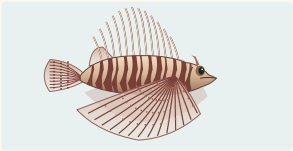
LEFT: original anatomy schematic; pectoral ray drawing is illustrative. RIGHT: actual Linux engine capture, shared texture maps and two stable palettes. The mesh includes 13 dorsal, 3 anal and paired pelvic spines. [1, 2]

01 · Model the silhouette

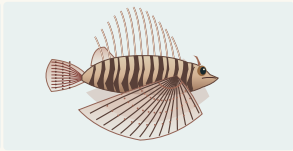
Deep head, tapering trunk, rounded tail and distinct soft fins. Keep the eye on the skull; open the jaw and throat rather than splitting the entire face. Use 13 tapered dorsal spines and fleshy head tabs. [1, 2]

IMPLEMENTED: 4,068 triangles and one visual actor per fish, with named regions and independent UVs. 13 dorsal, 3 anal and paired pelvic spines.

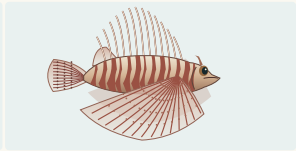
02 · Paint a surface, not rings



Wine / cream



Umber / sand

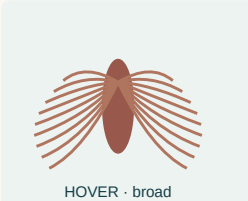


Chestnut / ivory

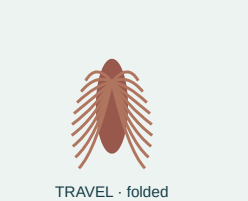
Uneven red-brown/cream body bands; spotted soft dorsal, anal and tail fins. [1] Separate texture coordinates from deformation weights.

IMPLEMENTED: same 256² body/fin maps for every fish; individual stable palettes, including future spawns. Pattern/UVs/alpha stay identical; use existing translucency for fin membranes.

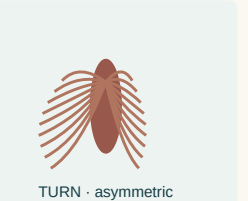
03 · Fins respond to the task



HOVER · broad



TRAVEL · folded



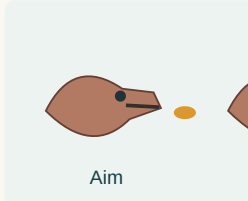
TURN · asymmetric

Field observations distinguish relaxed posture from travel with fins/spines folded back and more rapid tail strokes. [3] Smooth transitions; keep the roots stable and let curved tips lag.

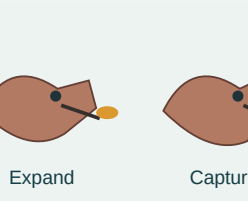
Observed mean speed	mm/s
Relaxed (15 events)	44.75
Traverse (12 events)	138.99
Strike (5 events)	625.44

Small field sample; physical speeds are context, not scene-unit settings or fixed fin frequencies. [3]

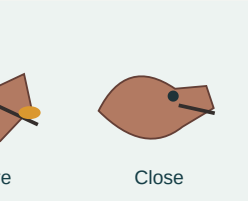
04 · One coordinated gulp



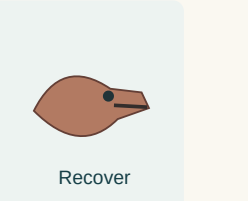
Aim



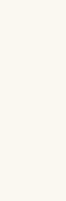
Expand



Capture



Close



Recover

Suction feeding combines gape, jaw rotation, head motion and hyoid depression. [4] Expansion drives a brief local flow: prey accelerates and can escape. Transfer ownership only on mouth entry; preserve inertia and off-axis convergence. No position tween.

GAME STARTING POINT: 0.26 s action / 0.10 s target are game choices. Actual capture requires aperture entry; misses remain possible. Review timing against footage.

2026 jet-blowing evidence concerns *P. miles*: useful context, not an automatic behavior rule for this *P. volitans* asset. [5]

05 · A small controller, a flexible mesh

IMPLEMENTED FORTH INTERFACE

```
\ phase · drive · turn · gulp
\ dark · light · actor
: lf-publish ( -- )
  tk-phase tk@ tk-pose-drive tk@
  tk-pose-roll tk@ .012 /
  gf-bite tk@ lf-palette
  650 tk@ lion-pose ;
```

Publish a compact pose packet; deform cached rest vertices in native code. Keep texture UVs intact. Each fish gets its own phase and turn bias.

Profile current → mesh/texture → deformation/feeding. Record script/native/render time, draw calls, actors, texture memory and p95 pacing on both Chromecasts. More triangles alone are not a realism or FPS measure.